Supplementary Material for Topology-Aware Differentiable Triangle-Soup Reconstruction via Persistent Homology

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A. The allocation-channel study, in brief

This appendix reports, with their numbers, the three findings of our allocation-channel study — controlled resampling-prior experiments on the same pipeline, scenes, and budgets, conducted before moving topology into the objective — that this paper builds on.

(1) Geometric metrics are topology-blind. Three constructed cases each pit a topologically correct candidate A against a wrong one B whose geometry is bisection-matched, making Chamfer equal by construction (40k samples, metrics normalized by the ground-truth bounding diagonal; Table S1). The discriminating-dimension bottleneck separates every pair — by 35–40× in the H0/H2 cases — while in the first two cases Hausdorff$_{95}$ actually prefers the topologically wrong candidate. Chamfer cannot tell a benign bump from a destroyed void; the persistence metric can.

(2) The prior arms, and why voids are "width-primary". The study biases DiffSoup's own resampler with a precomputed importance field built from the target's significant features. Each significant feature f (lifetime p_f above the significance threshold) contributes Gaussian kernels at the coordinates C_f of its birth/death-simplex vertices — the loop-filling triangle, void-filling tetrahedron, or component-merging edge — sized by the feature's own death scale:

$$\phi_d(q) = \sum_{f:\,\dim f = d} \; \sum_{c \in C_f} w_f \, e^{-\|q-c\|^2/(2\sigma_f^2)}, \qquad w_f = p_f/s^2,$$
$$\sigma_f = s\,\mathrm{clip}\big(\sqrt{\mathrm{death}_f},\, 3r_{\mathrm{med}},\, \tfrac{1}{2}\mathrm{diag}\big), \tag{1}$$

evaluated on a dense target-surface sample, normalized per dimension to $[0,1]$ (99.5th percentile), combined by max; a query takes its nearest surface sample's value. The spread knob s widens each kernel while preserving its injected surface mass ($\sigma \to s\sigma$, $w \to w/s^2$, so $w\sigma^2$ is s-invariant): s=1 is the concentrated prior (B2), s=3 the spread prior (B4) — the field C5 reuses. The field enters resampling at strict budget parity through two levers: prune protection — faces drop in order of vis + $\lambda\, Q_{0.9}(\mathrm{vis}) \cdot \mathrm{imp}$, removing exactly the required count — and respawn bias — top-importance visible faces subdivide first, then the protective order re-prunes to the cap. Remaining arms: B0 baseline; B1 = B2 applied at initialization only; B3/B5 random-center non-topological controls width-matched to B2/B4 respectively. On the void class (Table S2; N=1200, five paired seeds) B4 cuts the bottleneck 33–53% below baseline — but the width-matched random control B5 recovers most of that gain, leaving only a small, shape-dependent topological residual (cylinder 4.6σ, sphere 2.2σ, cube tie); one-shot placement washes out entirely (B1 ≈ B0, .0546 vs .0535 on the sphere). Hence the verdict quoted in the main paper's introduction: the prior's topological value is carried largely by its spatial width.

(3) No prior shape repairs loops. On the torus (N=700, five paired seeds, true β_1=2) every arm fails the topology-specificity test: the best condition is the non-topological wide control B5 (.0267), ahead of spread-topological B4 (.0316), baseline (.0409), and narrow random B3 (.0397); the concentrated topological field B2 is worst (.0425) and manufactures phantom handles — final significant-H1 count 4.4 against the true 2.0, with the worst Chamfer (1.32 vs baseline 1.01). Spreading merely stops the harm; it never beats the random control. That failure is the opening premise of the main paper, and the class the loss channel wins there (main paper §5.1, Table 1).

Table S1. Blindness probes: A topologically correct, B wrong, geometry bisection-matched so Chamfer is equal by construction. Bottleneck to ground truth in the discriminating dimension.

case (disc. dim)	Cham.% A = B	bott. A	bott. B	ratio
thin handle, genus 0→1 (H1)	0.916	≈0	.0302	—
bridge merges two components (H0)	0.628	.0014	.0574	40×
punctured enclosed void (H2)	0.977	.0040	.1385	35×

Table S2. Allocation study, void class (H2), N=1200, five paired seeds: tail bottleneck to target (lower is better). Spread beats concentrated; the width-matched random control (B5) recovers most of the spread gain.

shape	B0	B2 conc.	B3 narrow rand.	B4 spread	B5 wide rand.
sphere	.0535	.0438	.0440	.0357	.0374
cube	.0579	.0409	.0403	.0273	.0270
cylinder	.0545	.0440	.0477	.0360	.0402

Table S3. Significant bars (H1 / H2) on the clean target surfaces vs. sampling density M; floor = $6\,r_{\mathrm{med}}(M)$, seed 0.

	M=512	M=1024	M=2048	M=4096
sphere	0 / 1	0 / 1	0 / 1	0 / 1
torus	1 / 0	1 / 0	2 / 0	2 / 1
double torus	0 / 0	2 / 0	2 / 0	4 / 0

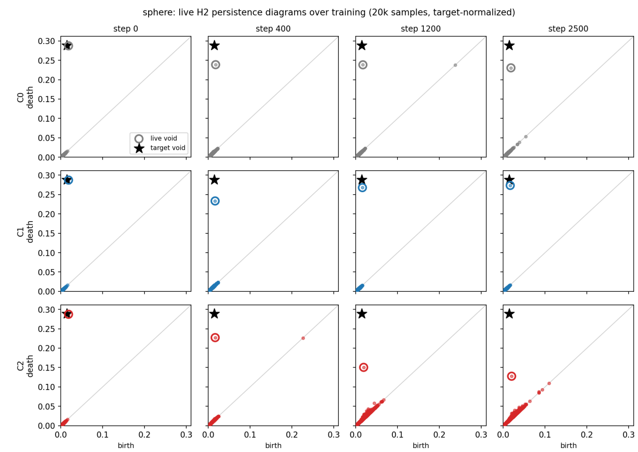

Figure S1. Live H2 persistence diagrams over training (one sphere seed; 20k samples, target-normalized; star = target void bar, circle = most persistent live bar). Top, baseline: the void drifts off target. Middle, loss: the void returns to the target once the ramp activates. Bottom, repulsion control: the void is pinned and dragged away while near-diagonal phantom bars accumulate.

B. Diagnostics: density sensitivity and diagram trajectories

Sampling density vs. measurability. Table S3 counts the significant bars of the clean target surfaces across sampling densities under the self-calibrating floor $6\,r_{\mathrm{med}}(M)$ of the main paper's §6. It confirms the three density claims made in the paper: the sphere's void clears the floor at every tested density; the torus needs M=2048 before its second loop is measurable, and its own void becomes measurable only at M=4096 (hence the H1 restriction at M=2048); and the double torus's tube loops cross the floor between M=2048 and M=4096 — consistent with the $M \gtrsim 3{\times}10^3$ estimate there — so all four loops are measurable at M=4096.

Diagram trajectories: the pinning mechanism, visualized. Fig. S1 shows the live H2 diagrams of one sphere seed under C0/C1/C2 at four training steps. All three arms start with the void essentially on target (initialization samples the SfM point cloud), and photometric training alone drags it away (the baseline settles ≈.06 below). The loss arm dips identically while the ramp is inactive, then pulls the void back to within ≈.016 of target — matching its measured tail. The control arm is qualitatively different: the void's death is dragged progressively down (≈.23 → .15 → .13) while a carpet of phantom mid-persistence bars accumulates near the diagonal — the repulsion-equilibrium signature discussed in the main paper's §6.

The genus-2 stretch case, in full. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over seeds), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the allocation study's floor, §A) and dominate both diagram metrics regardless of guidance. The loss adds no phantoms (#sig β_1 = 2 every run) at marginally best Chamfer.

Sensor-noise internals (C7/C7h). Recruitment absorbs the injected noise: at σ=1% the torus's second loop is sub-threshold at every refresh, with zero unreached targets. Calibration is the noisiest stage (λ_{peak}

Table S4. Tom-yum pot challenge inventory: what makes the shape hard, and the measured consequence in this study.

challenge	measured consequence
raw artist export (6,318 vertices): 9 open shells, mixed quad/12-gon faces, 50 boundary edges after exact weld, 232 non-manifold junction edges	no trustworthy ground-truth topology as exported; ingest-and-certify required (weld, offset-solidify into a thin shell via an exact unsigned distance field + marching cubes at iso ε, drop 804 enclosed pocket shells)
certified shell: one body, genus 3, β=(1, 6, 1) exact, 190,990 vertices	ground truth known by certificate (exact simplicial homology; watertightness cross-check; bit-identical rebuilds), not by construction
six thin handle/vent loops	every H1 loop sits below the significance floor at every working density — H1 is unusable as the loss observable
narrow flue mouth, capping at small α	the flue interior reads as an enclosed on-axis chamber — an H2 void, first measurable at M=8192 (margin 1.20–1.23×, five sampling seeds); the floor rule picks class and density prospectively (Table S5)
in-loop outcome	C1 cuts the void's error 3.9× at Chamfer parity; the repulsion control destroys it (β_2 1→0, all seeds, 1.36× worse Chamfer) — main paper Table 2

Table S5. Tom-yum pot staircase (seed 0; the M=8192 margin spans seeds 0–4): significant-bar signature ($\beta_0, \beta_1, \beta_2$) of the certified shell vs. sampling density M, floor = $6\,r_{\text{med}}(M)$. The bundle is built at the first density where the discriminating feature clears (M=8192: exactly one significant bar, H2, stable to re-sampling at $\sim 10^{-5}$).

	M=2048	M=4096	M=8192	M=20000
sig. bars	(1, 0, 0)	(1, 0, 0)	(1, 0, 1)	(1, 0, 1)

spreads ≈3× across seeds), yet tails stay tight — consistent with the flat ρ response reported in the main paper.

The tom-yum pot: challenge inventory and prospective floor protocol. The generality wave's artist mesh (main paper §4, §5.2) compounds the failure modes real scans will bring. Table S4 inventories them with the measured consequence of each — every row is backed by the ingest certificate or by a measurement in this study, none is asserted. Table S5 records the staircase behind the flue row: the certified shell's significant-bar signature across sampling density, which fixed the observable class (H2) and the bundle density (M=8192) before any training run.

C. Additional result figures and the repair gate

Fig. S2 shows the generality-wave training trajectories (main paper §5.2, Table 2); Fig. S3 the per-seed tail spread behind the sphere row of main-paper Table 1; Fig. S4 the torus significant-count check behind §5.1's zero-phantom-handles claim.

The stage-3a gate (main paper §5). Fig. S5 shows three of the four repair probes run before integration: the loss alone (no photometric term, Adam directly on point positions) must repair defective clouds. The predicted location-blind pathology of recruitment did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned), while cutting the 1-D bridge raises the recruited bar's death monotonically toward target — the runaway concentrates on the only profitable direction.

D. A decade of reconstruction, cut by optimization philosophy

The main paper's related work argues one sentence that this appendix substantiates in breadth: reconstruction methods of the differentiable decade, whatever their representation, optimize geometric objectives (photometric fidelity plus geometric regularizers) or steer sampling under such an objective — topology, when correct, is correct by construction, by extraction, or by accident, never because the training objective measured it. Table S6 tabulates representative methods under this cut, including the nearest concurrent topology-aware entries; the subsections survey each family and the adjacent literatures this paper draws machinery or discipline from.

D.1. Geometry-oriented: implicit fields

Occupancy and signed-distance learning deliver closed surfaces by construction [54, 61, 62], and a fitting stack matured around the level set: Eikonal regularization [25], surface rendering [60, 83], SDF-based volume rendering [79, 84], geometric priors [87], high-resolution and accelerated variants [51, 57, 80], meshing for deployment [85], and kernel-based point-cloud fitting [39]. Across all of them the training objective is photometric or geometric; the guarantee of consistent topology comes from the representation, its correctness (the right genus for the scene) is emergent, and the price is dense field evaluation plus post-hoc extraction — the cost regime the soup family exists to avoid.

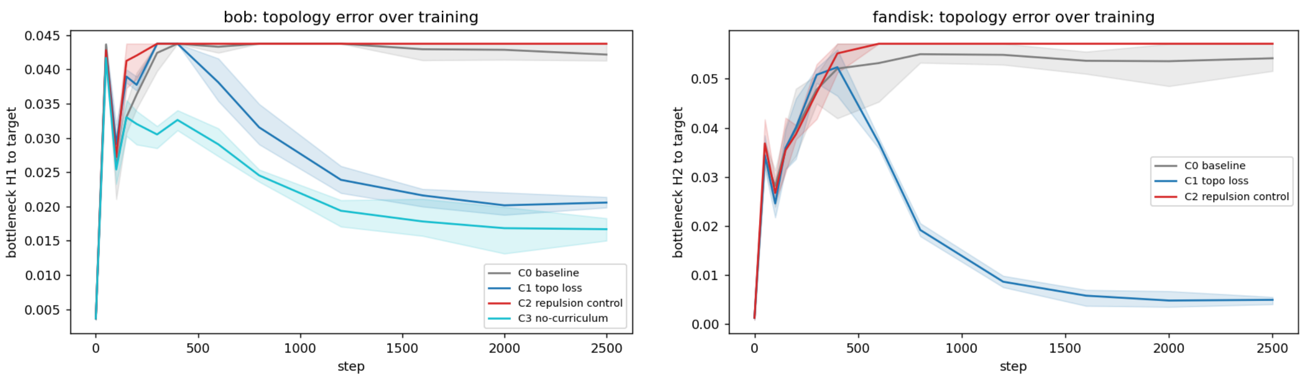

Figure S2. Generality trajectories (line = seed mean, band = seed min–max). Left: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with #sig H1 = 2 in every seed, while the control rides the baseline after collapsing a loop. Right: fandisk (CAD) — the largest effect measured in this study (10.4×).

Table S6. Representative reconstruction methods by optimization philosophy. "Topology in objective?" asks whether the training objective measures a topological quantity of the reconstructed surface (not whether the representation constrains topology, and not image-space summaries). Bottom block: the nearest topology-aware concurrent work (§D.7).

method	repr.	philosophy	topology in objective?	topology from
NeRF [55]	density field	geometry	no	emergent
OccNet [54]	occupancy	geometry	no	construction (closed iso)
DeepSDF [61]	SDF	geometry	no	construction (closed iso)
NeuS [79]	SDF	geometry	no	construction (closed iso)
Neuralangelo [51]	SDF	geometry	no	construction (closed iso)
Pixel2Mesh [78]	template mesh	geometry	no	frozen at template genus
Point2Mesh [30]	template mesh	geometry	no	frozen at template genus
Mesh R-CNN [24]	voxels → mesh	geometry	no	fixed by voxelization
DMTet [71]	grid + marching	geometry	no	extraction over the grid
nvdiffrec [58]	grid + marching	geometry	no	extraction over the grid
FlexiCubes [72]	grid + marching	geometry	no	extraction over the grid
Shape-as-Points [63]	points → implicit	geometry	no	construction (closed iso)
AtlasNet [26]	patch soup	geometry	no	unmeasured
DSS [86]	point soup	geometry	no	unmeasured
3DGS [47]	Gaussian soup	geometry	no	unmeasured
2DGS [37]	surfel soup	geometry	no	unmeasured
SuGaR [27]	Gaussians → mesh	geometry	no	post-hoc extraction
MILo [28]	Gaussians+mesh	geometry	no	in-loop extraction, unmeasured
Triangle Splatting [31]	triangle soup	geometry	no	unmeasured
Radiant Tri. Soup [8]	triangle soup	geometry	no	unmeasured (proximity forces)
Mini-Splatting [19]	Gaussian soup	sampling	no	unmeasured (budgeted resample)
DiffSoup [76]	triangle soup	geometry	no	unmeasured
DiffSoup + prior (§A)	triangle soup	sampling	no (steers only)	unmeasured; fails on loops
Topology-GS [73]	Gaussian soup	topology (image)	rendered images only	surface unmeasured
Gao et al. [22, 23]	watertight mesh	geometry	no (topology a prior)	template genus / camera prior
STITCH [41]	implicit (points)	topology	yes (β_0, live surface)	construction + constraint
this work	triangle soup	topology	yes (live-surface persistence)	measured, optimized, budgeted

D.2. Geometry-oriented: extraction, grids, and templates

Where the surface is explicit, its topology is decided by an extraction algorithm or frozen by a template. Marching Cubes and Dual Contouring fix per-cell connectivity by convention [43, 52]; Poisson reconstruction extracts an indicator iso-surface, watertight but with emergent genus [44, 45]; differentiable iso-surfacing makes extraction trainable without making topology a target [21, 58, 65, 71, 72]; learned tessellations move the per-cell decision into a network [14, 15]; and template pipelines fix topology up front and then only deform [24, 30, 78]. Robust meshing gives the same guarantee-by-construction for simulation inputs [36].

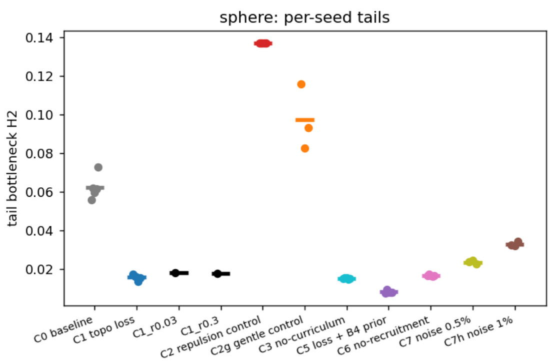

Figure S3. Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from above; the ρ ablation arms are indistinguishable from ρ=0.1.

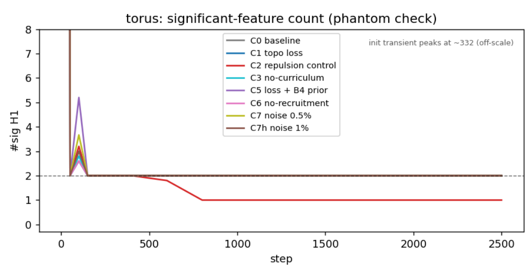

Figure S4. Torus phantom check (main paper §5.1): significant-H1 count over training (seed means; dashed line: true β_1=2). After the brief resampling transient near step 100, every loss arm sits exactly on two significant loops — zero phantom handles — while the repulsion control (C2) collapses a loop by step ~800 and never recovers it.

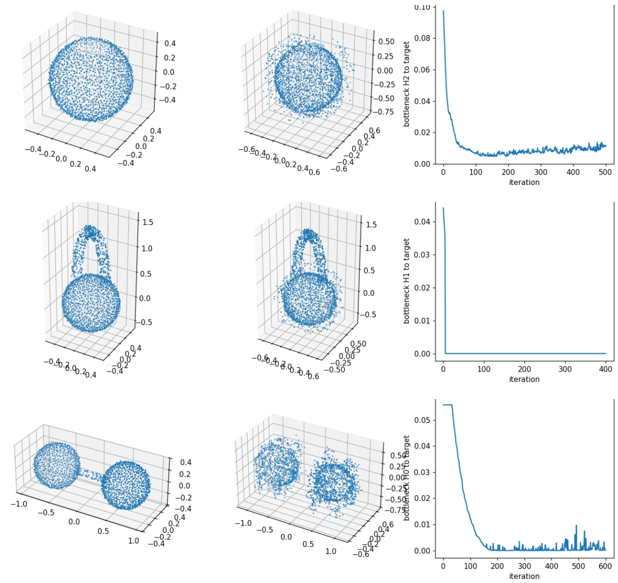

Figure S5. The stage-3a gate: the loss alone (Adam on raw points, no photometric term) repairs defective clouds; each row shows initial cloud, final cloud, and bottleneck-to-target trajectory. Top (T1): punctured sphere — the matched birth term closes the void (8.8×). Middle (T2): spurious handle — the diagonal term kills the extra H1 bar. Bottom (T4): bridged merge with no significant live H0 bar — recruitment severs the bridge (450×, β_0 1→2), the case where plain optimal matching provably has zero gradient. (T3, mis-spaced components, 1229×: omitted for space.)

In every case topology is an input or an artifact of the algorithm — measured nowhere, optimized never.

D.3. Geometry-oriented: point splatting and Gaussian soups

The connectivity-free line runs from EWA surface splatting [92] through neural point rendering [2, 67, 82] to the Gaussian-splatting era [47, 88] — Pulsar's abstract advertises the family's freedom from mesh "topology problems" explicitly [49]. Its surface-minded members regularize geometry ever harder — flattened surfels [18, 37], planar compression [13], opacity-field extraction [89], alignment-driven meshing [27] — and MILo now extracts a mesh differentiably at every training iteration [28]; triangle soups follow the same pattern [8, 31, 76]. Yet even with connectivity in the loop, no member of the family measures a topological quantity of what it builds: the objectives remain photometric fidelity plus geometric regularity.

D.4. Sampling-oriented: steering where a fixed objective spends

The second occupied family changes where primitives go, not what the objective asks of them: split-and-prune densification in 3DGS [47], budget-constrained Gaussian resampling [19], importance-driven respawn in DiffSoup [76] — and the topology-inspired allocation priors of our own study (§A), which put the family's implicit hypothesis to a controlled test and returned the null result the main paper builds on. Steering allocation is a real lever for geometric coverage; the study shows it is structurally the wrong lever for one-dimensional topology.

D.5. Topology as machinery: persistence in and around learning

The ingredients for topology-aware objectives have been maturing for two decades: persistence computation at scale [5, 53, 91], practical diagram metrics [46], stability theory [17], and vectorized representations [1, 7] on the computational side; trainable

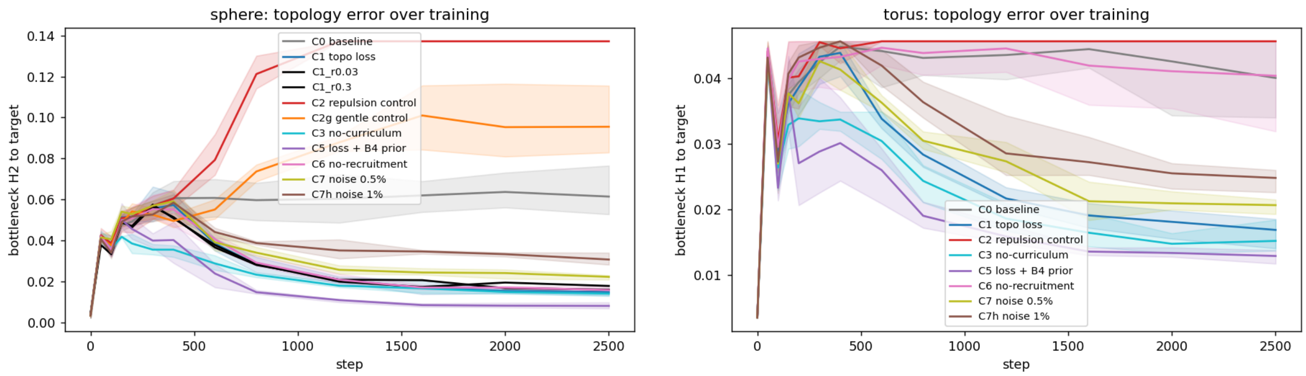

Figure S6. Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min–max). Left, sphere (H2): every loss arm (C1, C3, C5, C6, both ρ variants, noise pair C7/C7h) converges to a fraction of baseline; the norm-matched repulsion control C2 is destructive, and its gentle variant C2g still ends worse than baseline. Right, torus (H1), the class no prior won: same ordering, C5 best; noise arms descend shallower but never approach baseline; the no-recruitment ablation C6 rides it.

Figure S7. The tom-yum pot: an artist-authored Blender model (raw export: 9 open shells, 232 non-manifold junction edges) ingested and certified to a single genus-3 shell, β=(1, 6, 1) exact (main paper §4). Right: cutaway — the narrow flue mouth caps at small α, so the flue interior reads as an enclosed chamber: the on-axis H2 void the floor rule selects at M=8192 (all H1 loops sit below the floor).

diagram layers [9, 32], filtration learning [33], and general differentiability frameworks [10, 50, 59] on the gradient side (surveyed in [11, 12]). In-loop topological losses already win in adjacent domains — representation learning [56], tubular and topologically faithful segmentation [16, 35, 74, 75], microscopy volume prediction [77], point-cloud repair [6, 64] — and differentiable non-persistence signals exist too [66]. What had not happened is the transfer of this machinery into image-based surface reconstruction.

D.6. Topology outside the loop: repair and benchmarks

Before differentiable pipelines, topology was corrected after the fact: spurious-handle removal [29, 81], general mesh repair [3, 42, 70] (surveyed in [4]), and soup-to-manifold conversion [36, 38]; Thingi10K quantifies how pervasive soup-like defects are in the wild [90] — our tom-yum pot's raw export is a typical specimen. The evaluation culture matches: standard pipelines and benchmarks — COLMAP [68, 69], PMVS [20], DTU [40], Tanks and Temples [48] — score accuracy and completeness by point distances; none carries a topological-correctness metric, the blindness our Table S1 quantifies on controlled probes.

D.7. The empty cell, and its nearest neighbours

Concurrent work has begun to press on the gap from several directions, and the distinctions matter. Topology-GS [73] puts a persistence term inside Gaussian-splatting training — but on barcodes of rendered images, a topological-perceptual loss; the surface's diagram is never measured, and densification adds primitives rather than budgeting them. The high-genus inverse-rendering line [22, 23] reconstructs meshes whose genus is guaranteed by a Gauss–Bonnet-matched template or guides camera placement with homology computed once on a reference shape — topology as a prior, assumed rather than measured, with ground-truth genus required up front. STITCH [41] does measure a differentiable persistence quantity on a live implicit surface — enforcing a single connected component — but from point-cloud input: no images, no primitives, no budget. Topology also enters generation as conditioning features [34], again outside the reconstruction objective. Partition the three requirements — (i) a topological quantity of the evolving surface measured differentiably in the objective, (ii) image-based reconstruction through a renderer, (iii) a fixed primitive budget — and each neighbour holds at most one; none holds all three. That cell — the bottom row of Table S6 — is what the main paper fills.

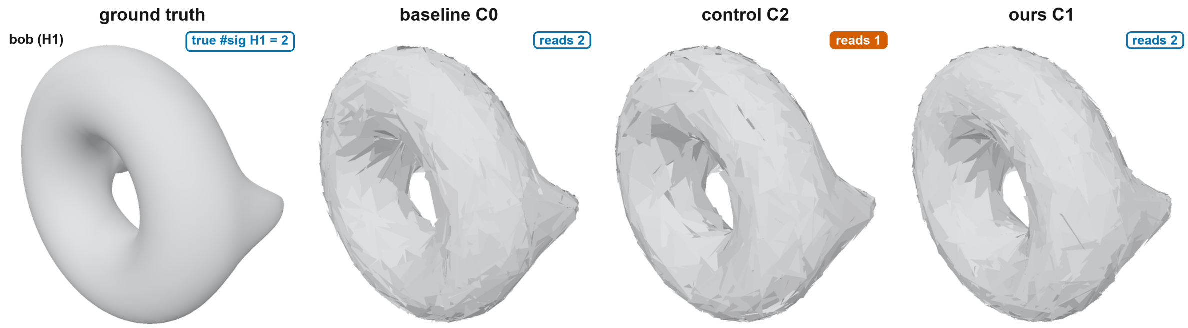

Figure S8. The loop-class row of the main paper's Fig. 1 matrix: bob (genus 1, H1). The repulsion control's reconstruction reads one significant loop against the true two (β_1 2→1, all seeds) at 2× worse Chamfer, while baseline and loss both read two — the loss at half the baseline's tail error (main Table 2).

E. Per-seed values and calibration sweeps

The main paper's reporting policy (§4) quotes Welch σ separations only where both arms ran five seeds; every three-seed verdict instead rests on disjoint per-seed ranges. Table S7 lists the underlying per-seed tail bottlenecks for the generality wave and the three-seed analytic shapes. In every row the largest C1 seed is well below the smallest C0 seed; C2's identical values across seeds are genuine (the repulsion equilibrium pins the diagram, main paper §6), not rounding.

Table S8 tabulates the two calibration-knob sweeps whose verdicts the main paper states in §3.4 and §5.1: the ρ response is flat over a 10× range, and the ramp window is immaterial — all alternatives land within seed noise of the default.

Table S7. Per-seed tail bottlenecks (seeds 0/1/2, run order) behind the three-seed verdicts. Spot's C1 value repeats across seeds at this precision; the pot rows are the quicklook values behind main Table 2.

shape (dim)	C0	C1 loss	C2 control
spot (H2)	.0429 / .0584 / .0550	.0237 / .0237 / .0237	.0652 ×3
bob (H1)	.0429 / .0413 / .0437	.0200 / .0207 / .0216	.0437 ×3
fandisk (H2)	.0540 / .0515 / .0559	.0058 / .0055 / .0042	.0571 ×3
tom-yum pot (H2)	.0218 / .0223 / .0223	.0059 / .0056 / .0056	.0223 ×3
cube (H2)	.0560 / .0551 / .0636	.0071 / .0064 / .0086	.1346 ×3
two spheres (H0)	.0060 / .0074 / .0062	.0007 / .0009 / .0005	.0009 / .0009 / .0006

Table S8. Calibration-knob sweeps (sphere, tail bottleneck). Top: ρ response, matched seed-0 sweep — the five-seed ρ=0.1 arm is main Table 1's C1 (.0156±.0013). Bottom: ramp-window pilot, three seeds each.

ρ (seed 0)	0.03	0.1	0.3
tail	.0181	.0172	.0177
ramp window	10–40%	20–50% (default)	30–60%
tail	.0154±.0017	.0156±.0013	.0179±.0019

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